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ADITYA SILVER OAK INSTITUTE OF TECHNOLOGY
DIPLOMA IN ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

A Report on

Lifynk

“A Verified Blood Coordination Platform”

The Smart HealthCare Technology (@ Viksit 2047)

Under Subject of

PROBLEM BASED LEARNING

&

Solution

Semester – IV

Submitted by

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**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



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1. Abstract

Blood availability during medical emergencies is one of the most critical challenges in India's healthcare system. The current process of finding compatible blood donors relies heavily on manual coordination through phone calls, WhatsApp groups, and social media platforms — all of which are unverified, slow, and unreliable. This results in fake donor requests, dangerous delays during the Golden Hour, and a complete absence of real-time communication between blood banks, hospitals, NGOs, and donors.

Lifynk is a centralized, real-time, verified blood coordination web platform designed to digitalize and streamline the complete blood donation and distribution workflow. The platform connects five interconnected pillars — Donors, NGOs, Blood Banks, Hospitals, and Recipients — through a single unified digital ecosystem.

Donors register on the platform and are guided to nearby NGOs, Blood Banks, and Camps through a live GPS-powered map. Blood Banks conduct full medical screening and store donor reports digitally as a Medical Resume, building a Verified Donor Pool. Hospitals access real-time blood stock across nearby Blood Banks and trigger emergency broadcasts when stock is unavailable, notifying the nearest pre-verified compatible donors instantly.

The platform is built on HTML, CSS, and JavaScript for the frontend, Firebase for OTP-based phone authentication, Firestore as the real-time database, Cloudinary for secure medical document storage, and deployed on Vercel. Lifynk eliminates fake requests through a multi-gate verification system, ensures compliance with India's DPDP Act 2023, and provides complete transparency and audit trails for all blood transactions across the network — making blood available to anyone, anywhere, in minutes.

HTML



Markup Language
Content

CSS



Style sheet Language
Presentation

JS



Programming Language
Behavior



Firebase

2. Introduction

2.1 Project Summary :

Field	Details
Project Name	Lifynk
Tagline	Linking Lives , Instantly
Theme	Smart Healthcare Technology
Type	Web Application
Domain	Medical Tech/ Smart Healthcare
Institution	ASOIT , Silver Oak University
Department	(Diploma) CSE – AIML , CSE – CS , BSC – Nursing
Guide	Prof. Naisargi Zala
Team	Rohan Burad , Akshay Singh , Hemang Barot , Ziya Sharma , Pallav Choudhary
Tech Stack	HTML/CSS/JS/Firebase/Firestore/Cloudinary ,Vercel

2.2 Purpose :

The purpose of Lifynk is to improve the connection between the 5 Pillars [Blood Banks, Donors, Hospitals, NGOs, and Recipients] during medical routine and emergencies.

The platform digitalizes the entire manual blood donation workflow — from donor registration and medical screening to emergency broadcasting via hospitals and blood delivery , Blood Donation Camps — into a single verified, real-time digital ecosystem. By providing each stakeholder a dedicated dashboard connected through one unified platform, Lifynk ensures that no patient loses their life due to a delayed or unverified blood request. We are here to develop a webApp with Zero Loopholes , due to the topic is very crucial , one step can risks the patients life.

2.3 Scope & Development :

The scope of Lifynk covers the complete digitalization of the blood donation and distribution lifecycle across five pillars:

- Development of a complete 5-pillar web-based e-Blood Bank platform.
- Donor registration with GPS-guided map showing nearby NGOs, Blood Banks, and Camps.
- Blood Bank digital screening report storage and Verified Donor Pool management.
- Digital Data Vault for Blood Banks to maintain, audit, and manage donor data and blood stock inventory.
- NGO camp scheduling, donor coordination, and reporting tools.
- Hospital emergency request system with two-step matching — checking available stock first, then searching the Verified Donor Pool.
- Multi-gate request verification system to eliminate fake requests.
- Critical Recipient real-time status tracking during emergency.
- Normal Recipient scheduled blood unit booking for conditions like Thalassemia.
- OTP-based phone authentication via Firebase for all stakeholders.
- Secure KYC and medical document storage via Cloudinary.
- Deployment on Vercel.

2.4 Objectives

- To design a centralized digital platform that connects all five pillars — Donors, NGOs, Blood Banks, Hospitals, and Recipients — in real-time.
- To ensure verification of all stakeholders through OTP authentication, KYC document upload, and NGO/hospital confirmation gates
- To reduce emergency response time using GPS-based location matching and a pre-verified Verified Donor Pool
- To provide Blood Banks with a Digital Data Vault to maintain, audit, and manage donor screening reports and blood stock inventory
- To digitalize the complete blood screening process — including ABO/Rh grouping, TTI screening, Hemoglobin estimation, Hematocrit, Antibody screening, and Nucleic Acid Testing — and store results as a donor's Medical Resume on the platform
- To serve both Critical Recipients (emergency situations) and Normal Recipients (Thalassemia/scheduled transfusions) through a single platform
- To eliminate fake blood requests through a multi-gate verification system
- To ensure compliance with India's Digital Personal Data Protection Act 2023

3. System Analysis

3.1 Study of Current System :

- The current blood donation and distribution system in India operates primarily through manual processes. When a patient requires blood urgently, the hospital or family contacts blood banks by telephone.
- If the blood bank does not have the required blood type in stock, the family resorts to posting on social media platforms like WhatsApp groups and Facebook, hoping that a compatible donor will respond.
- NGOs maintain their own independent databases of donors, which are not connected to hospitals or blood banks in real-time. Blood banks manage their inventory using paper registers or isolated software systems with no digital connection to other stakeholders.
- There is no centralized platform that brings all five pillars together under one verified, real-time ecosystem.

3.2 Problems in Existing system :

- **Manual Coordination Delays:** Families spend the Golden Hour — the critical first 60 minutes after a medical emergency — making phone calls and posting on social media instead of receiving blood
- **No Verification System:** Unverified donor and recipient information leads to fake blood requests, wasted donor time, and erosion of trust in donation networks
- **Data Silos:** Blood banks, hospitals, and NGOs operate in completely isolated systems with no real-time communication between them
- **No Donor Tracking:** There is no system to track when a donor is medically eligible to donate again, what donation type suits them, or how to alert them about nearby emergencies

- **No Screening Record:** Donor medical screening results are stored in physical files at blood banks with no digital record accessible to the donor or other authorized stakeholders
- **NGO Blindness:** NGOs planning blood donation camps have no data on where donors are concentrated, where demand is highest, or when to organize camps for maximum impact
- **Recipient Difficulties:** Thalassemia and other regular transfusion patients have no structured digital system to schedule and book blood units in advance

3.3 Requirement of New System :

- A centralized web platform connecting all five pillars — Donors, NGOs, Blood Banks, Hospitals, and Recipients — in real-time
- OTP-based phone authentication ensuring every account is tied to a verified real identity
- GPS-powered map guiding donors to nearest NGOs, Blood Banks, and Camps
- Digital Medical Resume for every donor — screening report stored on the platform, updated by Blood Banks after each screening
- Verified Donor Pool maintained by Blood Banks — pre-screened donors available for instant emergency matching
- Digital Data Vault for Blood Banks — complete donor data management, blood stock inventory, and audit trail
- Two-step emergency matching — check Blood Bank stock first, then search Verified Donor Pool
- Multi-gate verification for blood requests — Hospital/NGO/Family entry points with hospital confirmation gate.
- Real-time status tracking for families during emergency requests
- Pre-Blood Unit booking for Normal Recipients. Compliance with India's DPDP Act 2023 for sensitive medical data protection

4. Methodology

4.1 Requirement Analysis :

The requirements for Lifynk were gathered through a structured analysis of the existing blood donation ecosystem, literature review of government health portals (e-RaktKosh, NHM), and consultation with medical team members. Requirements were classified into:

- **Functional Requirements:** Donor registration, GPS matching, screening report storage, emergency broadcasting, hospital stock management, NGO camp scheduling, recipient blood booking.
- **Non-Functional Requirements:** System availability during emergencies, OTP authentication response time, data privacy compliance, role-based access control.
- **Medical Requirements:** Blood compatibility matrix (ABO/Rh), donor eligibility rules, screening test protocols, cooldown periods for each donation type.

4.2 System Design :

The system is designed as a 5-pillar architecture:

- **Donor Dashboard:** Registration, GPS map, Medical Resume, Nearby Requests, Hall of Heroes.
- **NGO Dashboard:** Camp scheduling, donor coordination, reporting engine.
- **Blood Bank Dashboard:** Donor screening management, Digital Data Vault, blood stock inventory heatmap.
- **Hospital Dashboard:** Emergency request system, blood stock viewer, donor matching results.
- **Recipient Interface:** Emergency request with status tracking, scheduled blood unit booking.

4.3 Hardware Requirements :

Component	requirement
Client device	Any Modern smartphone or computer
browser	Google Chrome v100+ , Firefox , Safari
internet connection	Minimum 2G/3G for basic functionality
server	Cloud – based – no dedicated hardware required
storage	Managed via Firebase and Cloudinary cloud services

4.4 Software Development :

Technology	role
HTML / CSS / JavaScript	Frontend - all dashboard interfaces.
Firebase authentication	OTP-based phone number verification.
Cloud firestore	Real-time NoSQL database.
cloudinary	Secure storage for KYC and screening documents.
Vercel	Deployment and hosting platform.
Leaflet.js / open street map	GPS map for Donor location progress.

4.5 Algorithm Development :

Two step Emergency Matching Algorithm :

Step 1: Hospital raises emergency blood request

➤ Input :Blood type required , units needed , hospital GPS location.

Step 2: Check nearby Blood Bank stock

- Query Firestore for Blood Bank within X km radius.
- Filter by required blood type and available units.
- If the stock found – Triggers Blood Bank to Hospital transfer.

Step 3: If no stock available – Search Verified Donor Pool

- Query donors by ABO/Rh compatibility matrix.
- Filter by GPS proximity to requesting hospital.
- Filter by eligibility status (Cooldown period completed).
- Rank by distance – Notify top 5 nearest compatible Donors.

Step 4: Donor accepts – Status updated in real – life

- Family sees : Pending – Searching – Match Found – Blood on way – Completed .

Whole Blood: 56 days cooldown | max 6 times/year

Platelets: 7 days cooldown | max 24 times/year

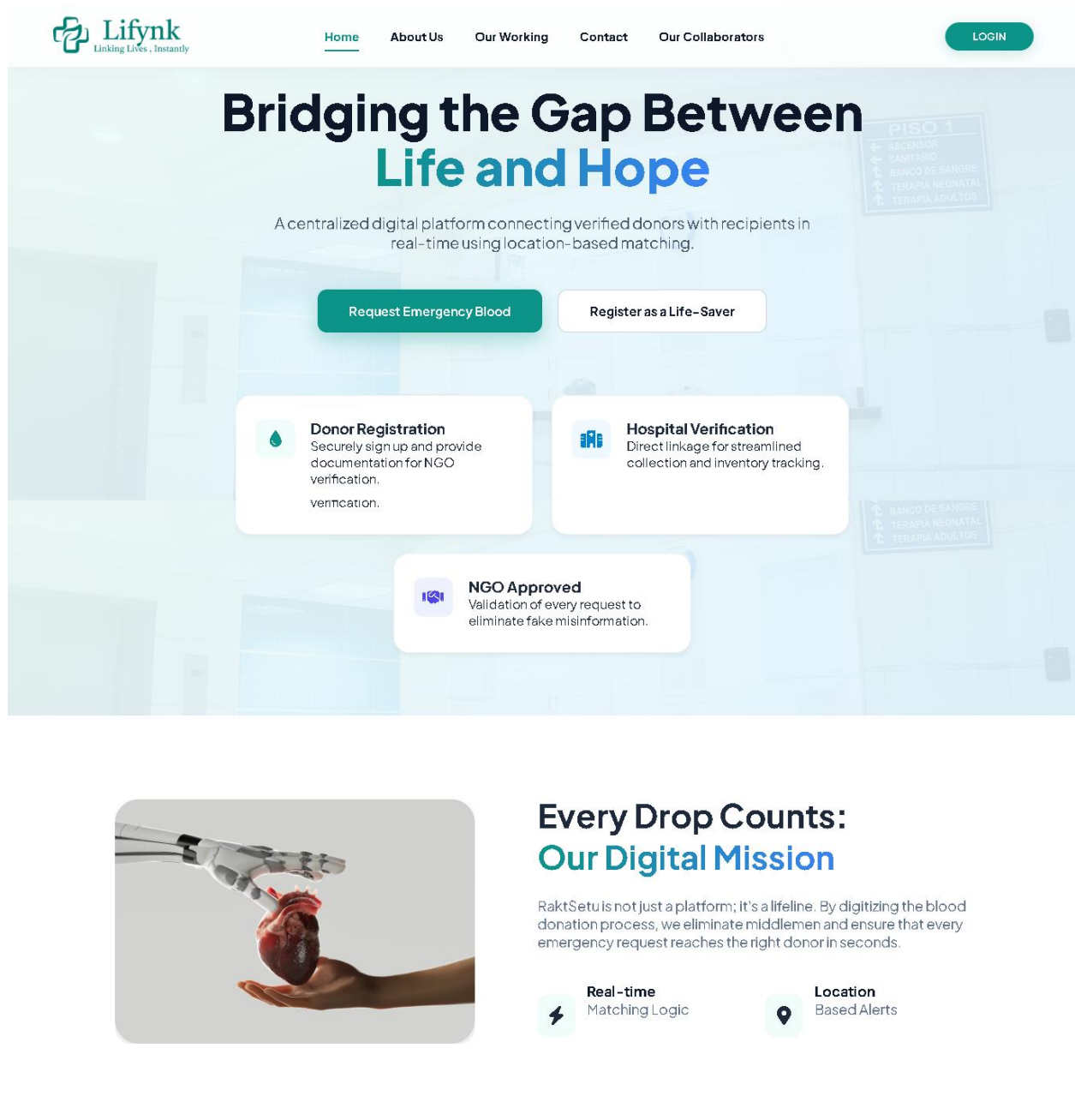
Plasma: 28 days cooldown | max 13 times/year

Double Red Cells: 112 days cooldown | max 3-4 times/year

6-month checkup popup → donor updates health data → Verified Donor
Pool record refreshed

5. PHOTOS

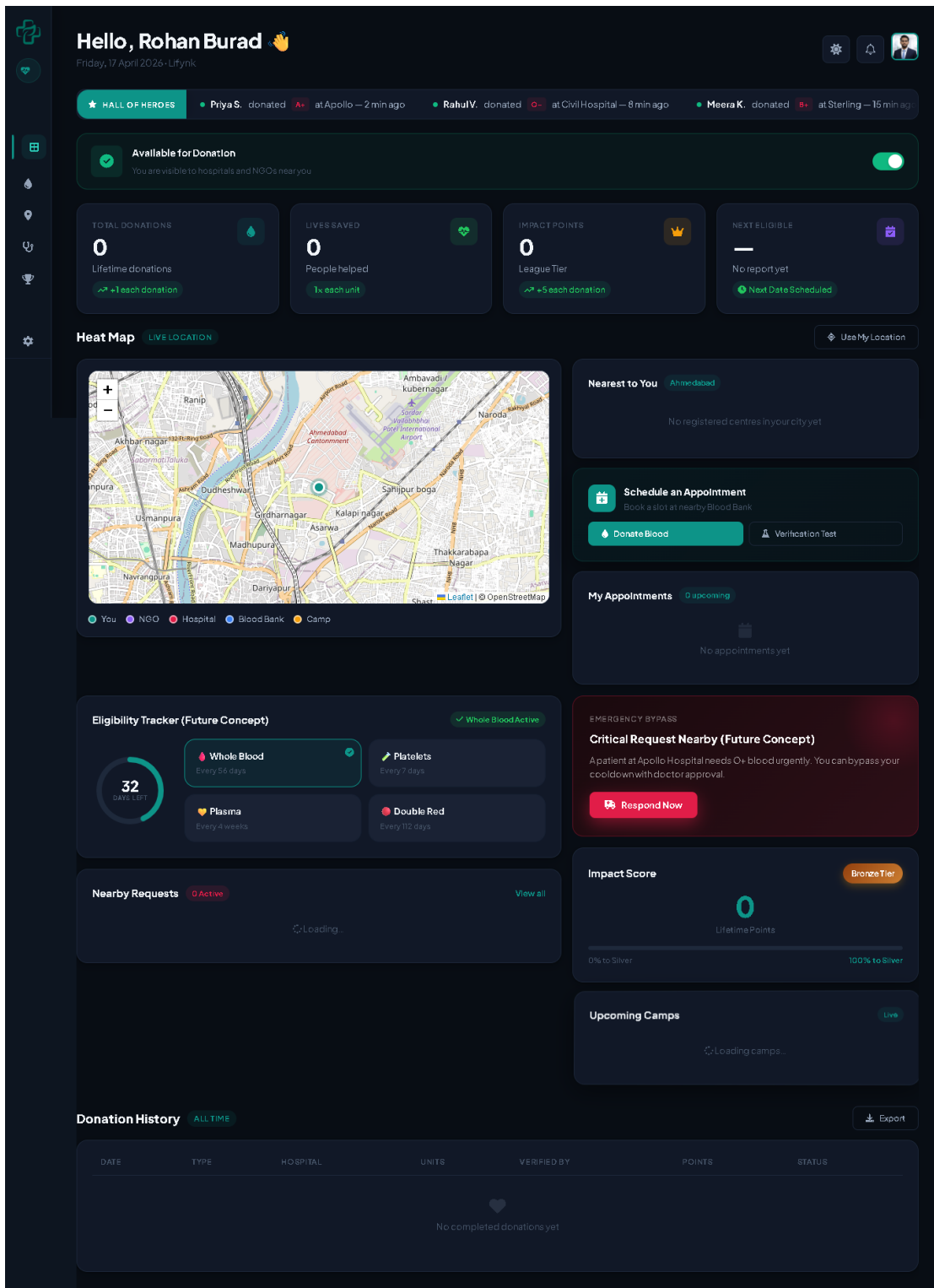
5.1 Landing Page of our website :



Terms & Conditions Apply

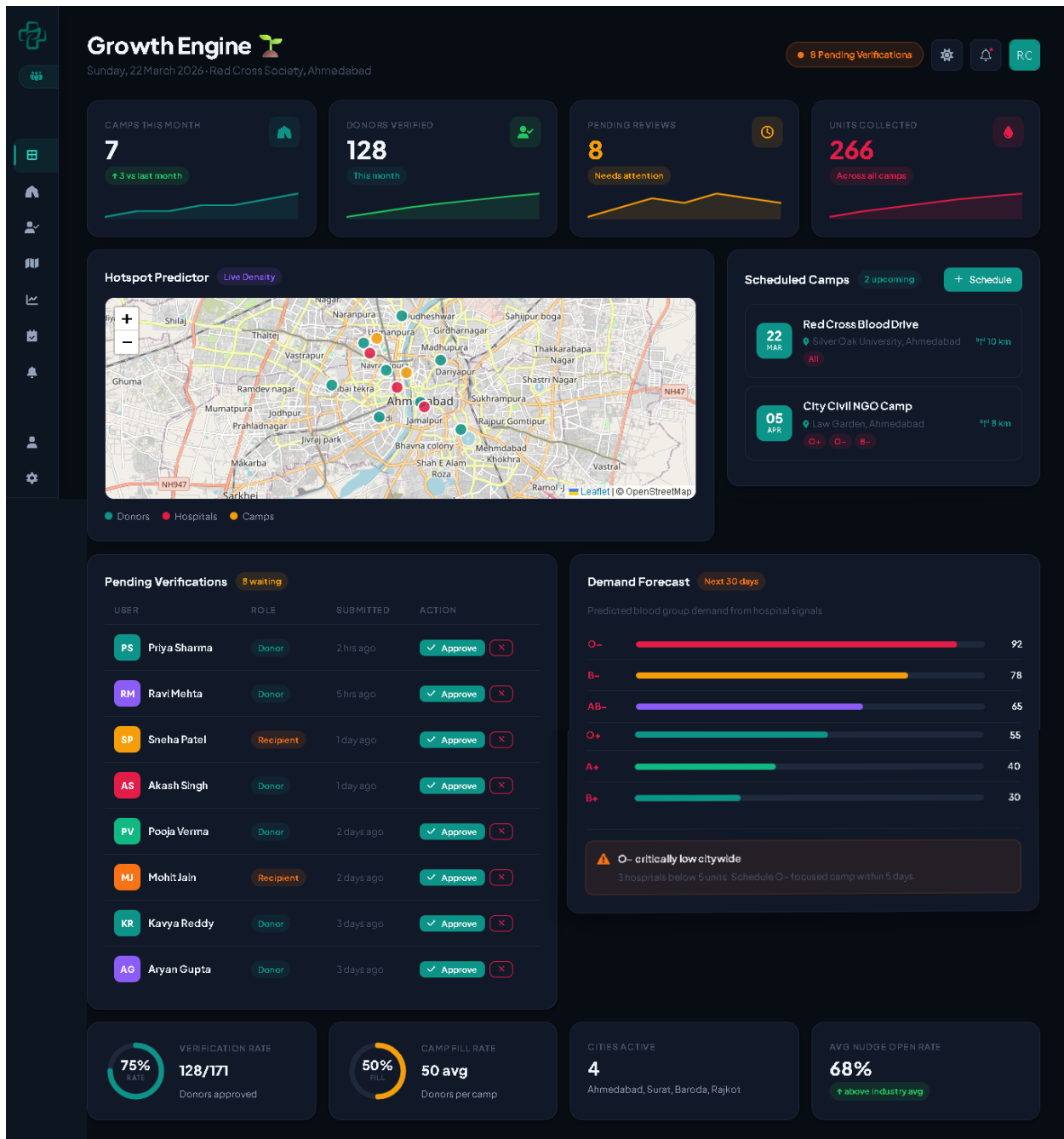
© 2026 Lifynk | Blood Donation Coordination System | All rights reserved.

5.2 Donors – Dashboard :



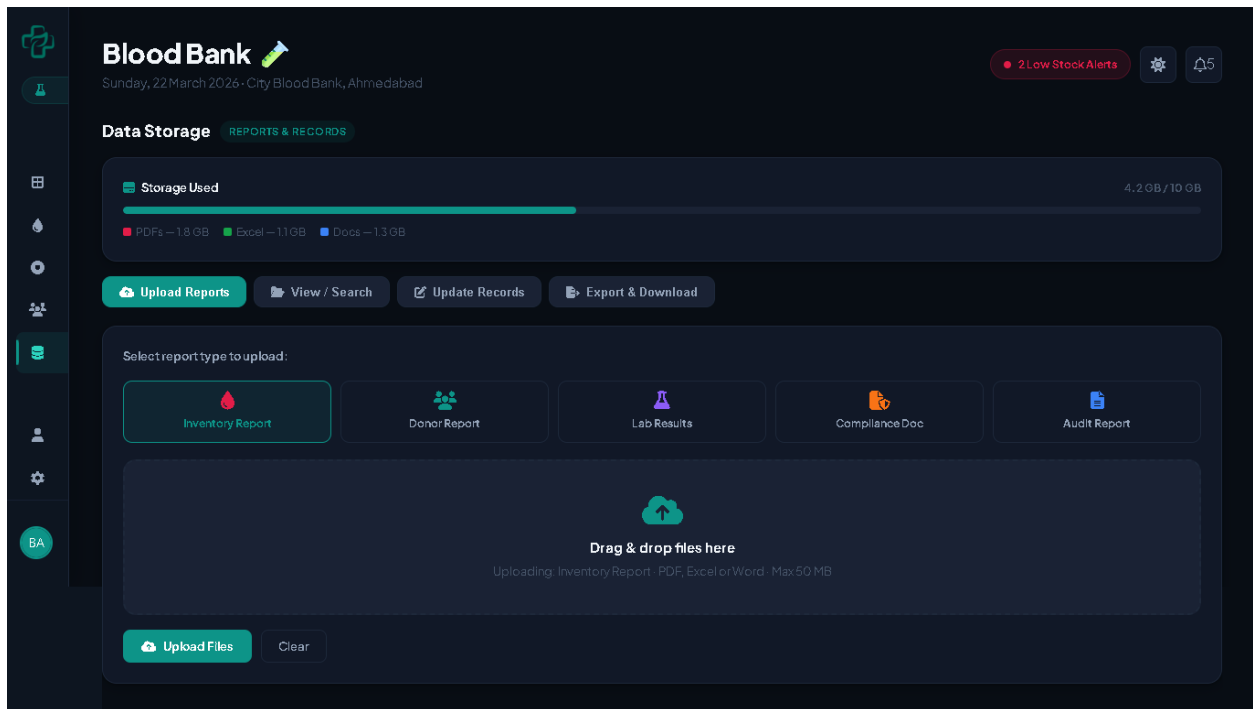
This is the donors's dashboard where donor's can use our services.

5.3 NGOs – Dashboard :



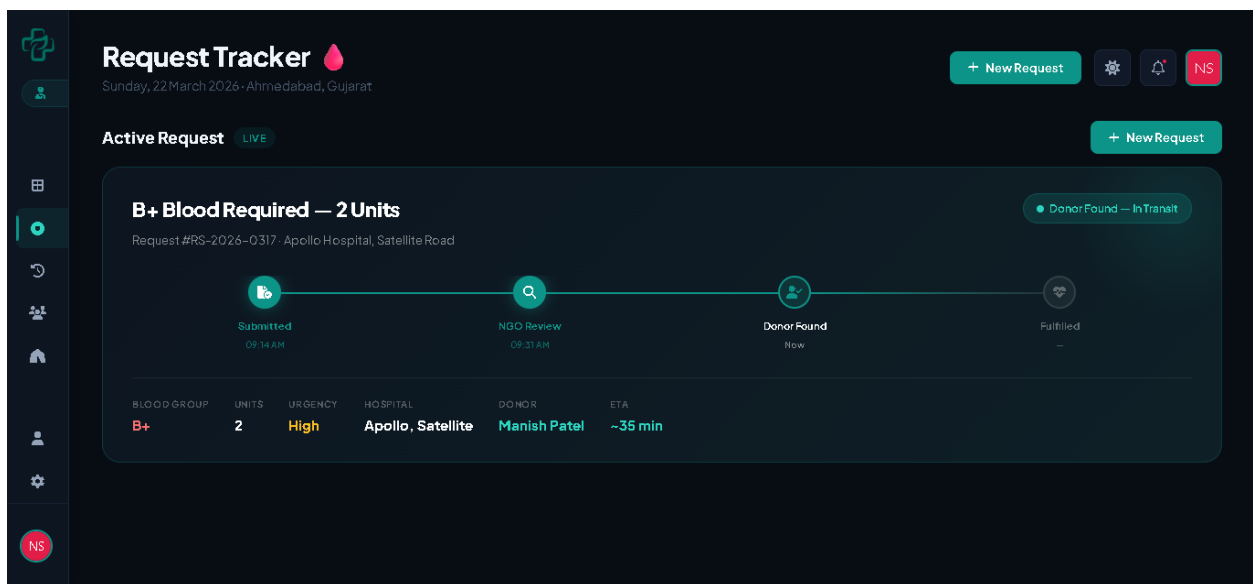
NGO : They have the ability to schedule camp and a Heat Map provided to see the scheduled camps , the nearby Hospitals and Donors.

5.4 Blood Banks – Dashboard :



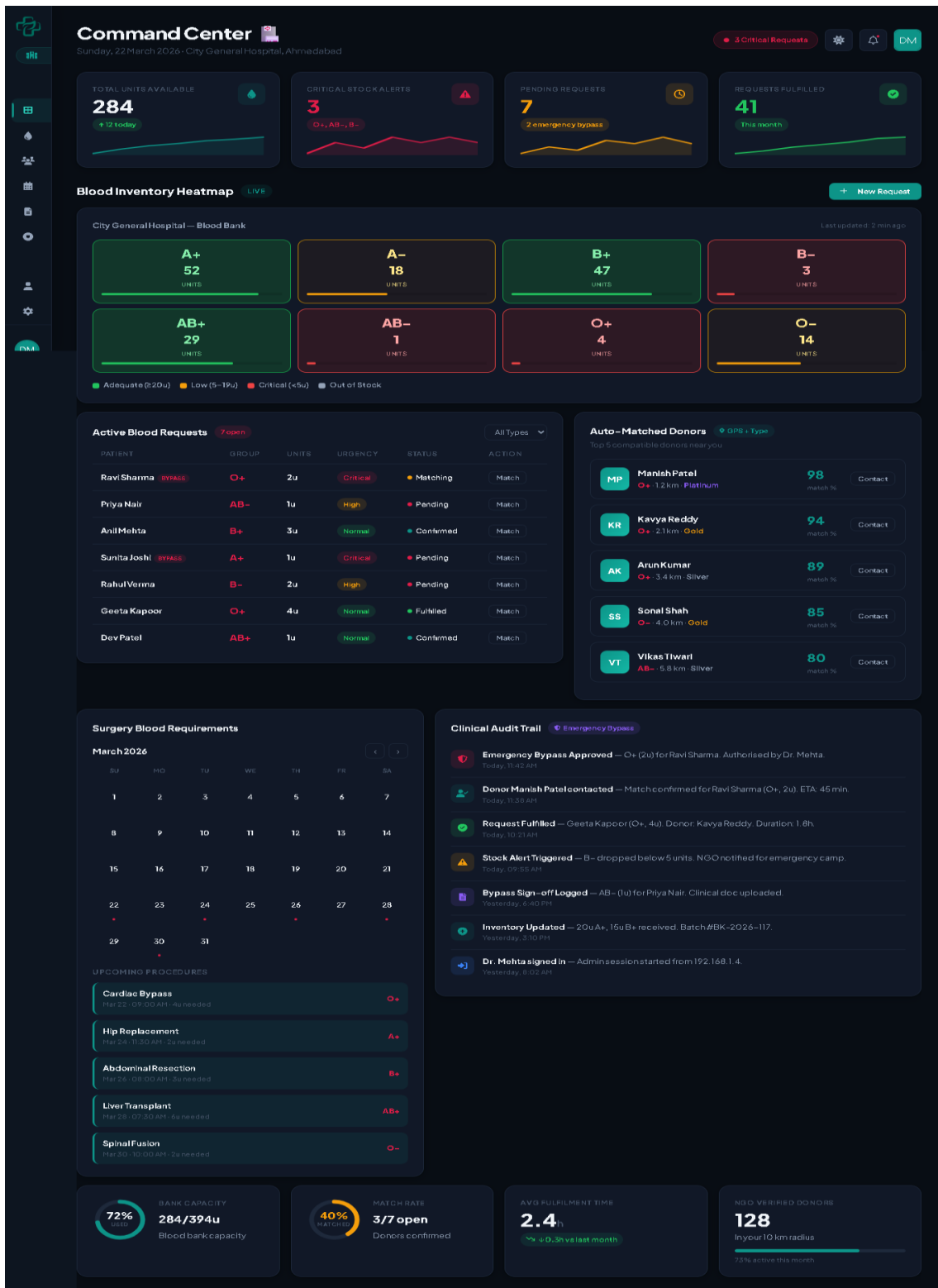
Blood bank can handle their inventory stocks and ledger digitally.

5.5 Recipients – Dashboard :



Recipient can track their pre booked blood units with our tracking systems

5.6 Hospitals – Dashboard :



6. Limitations & Future Enhancements

Limitations :

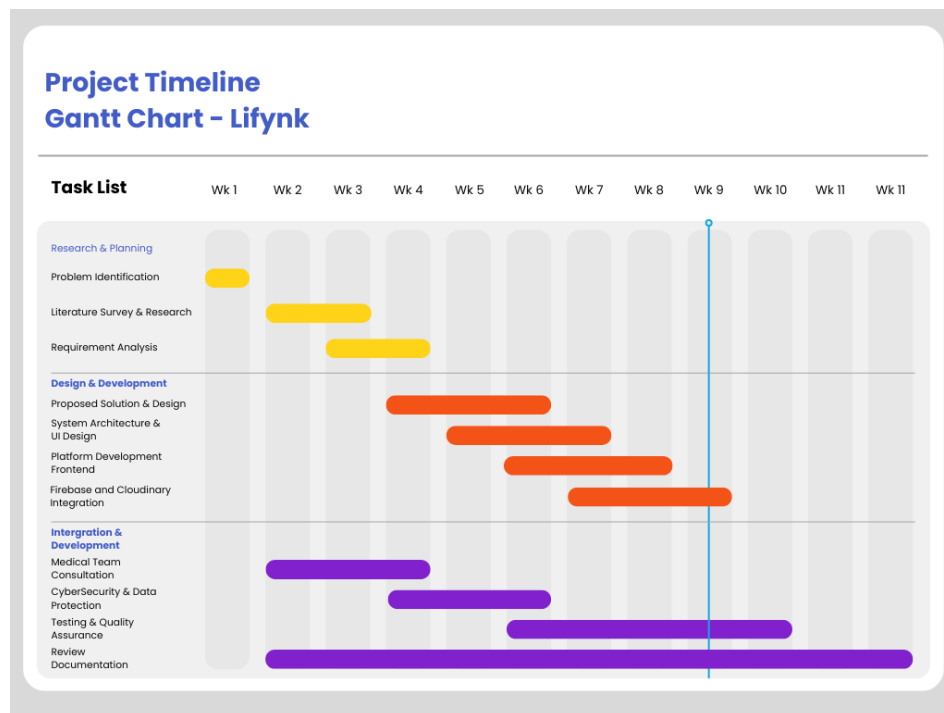
- Platform effectiveness depends on active registration and participation of all five pillars.
- Requires internet access and basic digital literacy from all stakeholders.
- Initial implementation will have limited geographic coverage — expanding over time.
- Blood Bank screening process still happens physically — platform only stores and manages the data digitally.
- Emergency response time depends on donor availability and willingness to respond.
- No mobile app currently — web platform only.

Future Enhancements :

- Native mobile application for Android and iOS for wider accessibility.
- AI/ML powered Geospatial Hotspot Predictor — identifies optimal locations for NGO camps using donor density clustering.
- AI-driven Blood Demand Forecasting — predicts hospital blood requirements based on seasonal patterns and surgery schedules.
- Integration with Government Health Systems — e-RaktKosh, National Health Mission.
- Real-time blood inventory API linkage with hospital LIMS systems.
- Automated donor health alerts and iron deficiency warnings for frequent donors.
- Multi-language support — Gujarati, Hindi for regional accessibility.

- Expansion to pan-India coverage across multiple cities and states.
- Gamification expansion — Hall of Heroes leaderboard, LinkedIn-shareable NGO-verified donation certificates.

7. Timeline Chart – Lifynk



8. Platform Development Roadmap

Research & Development

8.1 Completed Pillars (Live) :

The following dashboards are fully built, tested, and functional in the current version of the Lifynk Platform :

[I] Donor-Dashboard – Live

Completed Features :

- Account registration with Firebase OTP phone authentication.
- Live GPS map showing nearby Hospitals, Blood Banks, NGOs and Camps (Leaflet.js + OpenStreetMap).
- Nearest to Donors list – Showing name , type and distance in km.
- Overview metrics – Lives Saved , Impact Points, Next Eligible date and with the Blood Donation & Verification Test Schedule Section which helps to get the verified donors on our platform.
- My Donations - donation history per donation type.
- Medical Resume – screening report uploaded and managed by Blood Bank.
- Nearby Request – emergency requests visible to donor based on location.
- Hall of Heroes – gamification leaderboard.
- By using the jsPDF scripts now we can also download the system generated reports also if needed.

Verified Flow :

Donor registers → OTP verified → Get's Dashboard

Guided to nearby NGO/Blood Bank via map →

Screening done at Blood Bank →

Report stored as Medical Resume →

Donor enters Verified Donor Pool →

Receives emergency notifications → Responds to the emergencies.

[II] Blood Bank-Dashboard – Live

Completed Features:

- Blood Bank account registration and KYC verification.
- Donor Screening report upload and management.
- Blood stock inventory management per blood type.
- Verified Donor Pool Management.
- Ability to update donor Medical Resume after each screening.
- View incoming hospital blood requests.
- Respond to emergency donor match requests.

Verified Flow :

Blood Bank registers → KYC verified →

Receives donor for screening →

Conducts full screening (ABO/Rh, TTI, Hb, Hematocrit, Antibody, NAT) →

Uploads report to platform as donor's Medical Resume →

Donor enters Verified Donor Pool →

Manages blood stock inventory →

Responds to Hospital emergency requests

8.2. Pillars Under Active Research & Development :

The following dashboards are currently under active research and development. Flows are being discussed, validated with the medical team, and finalized progressively.

8.2.1. Hospital Dashboard – Under R & D

Current Research Status:

- Core UI structure designed.
- Emergency request raising flow in design phase.
- Blood stock visibility across nearby Blood Banks under discussion.

Flows Being Discussed:

- How hospital raises emergency blood request.
- Two-step matching logic — stock check first, then Verified Donor Pool search.
- How hospital confirms requests raised by Family or NGO.
- What data hospital sees from nearby Blood Banks — availability and distance only, not full donor details.

Open Questions / Design Decisions Pending:

- Should hospital see donor names during matching or only blood type + distance?
- What is the maximum radius for the initial stock search before expanding?
- Should the hospital dashboard have a surgery calendar for advance blood demand forecasting ?
- How does hospital mark a case as completed after blood is received?

R&D Notes: *(To be filled after medical team consultation)*

8.2.2. NGO Dashboard – Under R&D

Current Research :

- NGO registration and KYC flow designed.
- Camp scheduling module in planning phase.
- Donor coordination tools under discussion.

Flows Being Discussed:

- How NGO organizes and schedules blood donation camps
- How NGO guides donors to Blood Banks after camp collection
- How NGO raises emergency blood requests on behalf of families
- How collected blood samples from camps are channeled to Blood Banks

Open Questions / Design Decisions Pending:

- What data does NGO see from Blood Banks — full inventory or only shortage alerts?
- Should NGO be able to see the Verified Donor Pool or only their own camp donors?
- How does NGO manage and assign volunteers to donation camps?
- What reports can NGO generate for government submission?

R&D Notes: *(To be filled after medical team consultation)*

8.2.3 Recipients Interface – Under R&D

Current Status :

- Two recipient types identified — Critical and Normal
- Critical recipient emergency flow being finalized
- Normal recipient booking flow in design phase

Flows Being Discussed :

Critical Recipient:

- Three entry points for raising emergency request — Hospital Admin, NGO, Patient Family.

- Hospital confirmation gate for Family and NGO requests.
- Real-time status tracking — Pending → Searching → Match Found → Blood on Way → Completed.

Normal Recipient :

- Recipient registers and gets verified on platform .
- Books scheduled blood units directly from Blood Bank .
- Blood Bank confirms and assigns slot .
- Hospital looped in only when transfusion needs medical supervision.

Open Questions / Design Decisions Pending:

- Should Critical Recipient family see donor details during tracking or only status?
- What verification documents does a Normal Recipient need to submit?
- Should Normal Recipient booking have a recurring schedule option for Thalassemia patients who need blood every 3-4 weeks?
- How does the system handle a Normal Recipient booking when the required blood type is out of stock?

R&D Notes: *(To be filled after medical team consultation)*

8.3 Zero Loophole Flow Matrix

This table maps every critical user action to its expected outcome, known edge cases, and current resolution status. Updated progressively after each R&D and medical team session.

#	User	Action	Expected Outcome	Edge Case	Resolution Status
1.	Donor	Registers on platform	OTP Verified, account created	Donor uses fake number	Firestore OTP blocks unverified numbers
2.	Donor	Visits Blood Bank for screening	Full screening done, Medical Resume created	Donor is medically ineligible	System marks donor as ineligible, blocks from Verified Pool

3.	Donor	Cooldown period active	Cannot donate, timer shown	Emergency situation arises	Emergency bypass under R&D discussion
4.	Donor	6-month checkup popup appears	Donor updates health data, pool record refreshed	Donor ignores popup	Donor flagged as "Needs Update" in Blood Bank dashboard
5	Blood Bank	Uploads screening report	Report saved as Medical Resume, donor enters Verified Pool	Incomplete test results	System flags report as incomplete, donor not added to pool
6	Blood Bank	Updates blood stock inventory	Live inventory updated on platform	Stock expires before use	Expiry alert system — under R&D
7	Hospital	Raises emergency request	Platform searches nearby Blood Bank stock	No stock found within radius	System automatically searches Verified Donor Pool
8	Hospital	No stock AND no available donor	Request escalates	Radius expands, NGOs notified	Escalation radius logic — under R&D
9	Family	Raises emergency request	Goes to Hospital dashboard for confirmation	Hospital doesn't respond within X mins	Auto-escalation timer — under R&D
10	NGO	Raises emergency request	Goes to Hospital dashboard for confirmation	Hospital confirms	Request goes live on platform
11	Normal Recipient	Books blood units from Blood Bank	Booking confirmed, slot assigned	Required blood type out of stock	Recipient notified, alternate Blood Bank suggested — under R&D
12	Normal Recipient	Recurring Pre-booking of Blood Units	Auto-scheduled every X weeks	Blood Bank capacity full on scheduled date	Rescheduling flow — under R&D
13	Any User	Attempts to access another user's data	Permission denied	IDOR attack attempt	Firebase security rules block unauthorized access
14	Any User	Submits fake blood request	Request goes through confirmation gate	Confirmed fake by hospital	Account flagged, blacklist system — under R&D

9. Conclusion

Lifynk — "A Verified Blood Coordination Platform" — represents a comprehensive solution to one of India's most critical healthcare challenges: the unavailability of verified, real-time blood coordination during medical emergencies. By digitalizing the complete blood donation workflow across five interconnected pillars — Donors, NGOs, Blood Banks, Hospitals, and Recipients — the platform eliminates the manual delays, fake requests, and data silos that currently cost lives every day.

The platform's core innovation lies in its Verified Donor Pool and Digital Data Vault — a pre-screened, GPS-indexed database of donors maintained by Blood Banks that enables hospitals to find compatible blood within minutes rather than hours. Combined with a multi-gate verification system, real-time status tracking for families, and scheduled booking for regular transfusion patients, Lifynk addresses both emergency and non-emergency blood requirements through a single unified platform.

Built on modern web technologies with Firebase, Firestore, Cloudinary, and Vercel, the platform is scalable, secure, and compliant with India's DPDP Act 2023. As a Smart HealthCare initiative aligned with Viksit Bharat 2047, Lifynk demonstrates how technology can bridge critical gaps in India's public health infrastructure — making blood available to anyone, anywhere, in minutes.

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